

# Lie Groups of Symmetry and Their Application in Solving ODEs

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## Abstract

Ordinary differential equations (ODEs) are critical tools for mathematicians, engineers, or even social scientists to model dynamical systems, and on many occasions one need to solve for a “general solution” for a given ODE. However, students in an introductory ODE class often struggle to intuitively understand a variety of ad hoc methods tailored to solve only some specific types of ODEs. This expository paper introduces the Lie group theory as a more unifying way to craft general solutions to ODEs. After providing some background knowledge in both group theory and ODEs, I will first introduce the concept of one-parameter Lie groups of point symmetries, their infinitesimal generators, how they look like in an ODE (i.e. the symmetry condition), and how to reconstruct Lie symmetries by leveraging the symmetry condition. A big portion of this paper is devoted to developing canonical coordinate and using it to solve any first-order ODEs. In addition, I will briefly discuss Lie group’s application in high-order ODEs.

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## Introduction

In an undergraduate differential equation class, it is common that many beginners are confused over a range of ad hoc methods to solve a very narrow set of ordinary differential equations (ODE), such as the integrating factor and change of variables. This phenomenon had been the state of the art until the mid-nineteenth century, when a Norwegian mathematician called Sophus Lie made the remarkable discovery that all those tailored methods of solving ODEs were actually just some special cases of a general integration procedure. What is more fascinating is that such procedure heavily borrows from the language of abstract algebra, including group and group actions, which usually appear in a more discrete setting. These groups of great avail in sloving ODEs are continuous transformation groups composed of symmetries, more formally known as the Lie groups.

Synoptically speaking, in our application in first-order ODEs those groups “act” on the variable space  $(x, y)$ , namely the Euclidean domain  $\mathbb{R}^2$ , with a parameter  $\varepsilon$  that varies continuously and traces out “trajectories”, known as the orbits, of solution curves. As discussed in the following pages, the tangent vectors of the orbits are critical objects that allow us to first generate the Lie group and to construct a new coordinate system, known as the canonical coordinate, in which a first-order ODE can always be reduced to a quadrature (i.e. can be directly solved by integration techniques). In the case of higher orders of ODEs, Lie groups of symmetries can also symplify the solution by reducing the order of differential equations. The algebraic tools used in this case are fairly direct extension of those used in variable space  $(x, y)$  into the space  $(x, y', y'', \dots y^{(n)})$ .

This expository paper explores and summarizes Lie's much more universalizing way of finding general solutions to ODEs, aiming to popularize such content to a wider readership with some level of exposure to both differential equations and abstract algebra. Before diving into Lie's theory and its application, we provide a brief account of preliminary knowledge in both abstract algebra and differential equations that might help the readers traverse through the later sections more smoothly. The remaining pages progresses as the following: chapter 3 introduces and formulates the Lie group of point symmetry; chapter 4 dives into the symmetry condition of first-order ODEs and how Lie's method can solve any first-order ODEs up to a quadrature; chapter 5 explores ODEs of higher orders and discusses the possibility of order reduction. Finally, we end with a review of important results and some concluding remarks.

## Preliminaries and Background

### 1 Group Theory: basics

#### 1.1 Definition of a Group

**Definition 1.** A group is an ordered pair  $(G, \star)$ , where  $G$  is a set admitting a binary operation  $\star$  on  $G$  that satisfies the following properties [2]:

- i. The operation  $\star$  is associative, i.e.  $\forall a, b, c \in G, (a \star b) \star c = a \star (b \star c)$
- ii. There exists an identity element  $e \in G$  such that  $\forall g \in G, g \star e = e \star g = g$ .
- iii. For every  $g \in G$ , there exists an inverse  $g^{-1} \in G$  such that  $g \star g^{-1} = g^{-1} \star g = e$ .

**Note.** If the operation  $\star$  commutes on  $G$ , namely  $\forall a, b \in G, a \star b = b \star a$ , then  $(G, \star)$  is an *abelian*, or commutative, group.

**Example 1.** Below I provide some simple motivating examples of groups:

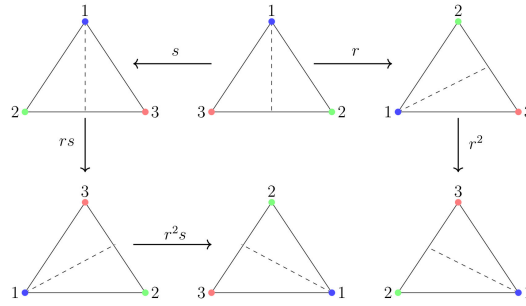
- i. The integers form an additive group:  $(\mathbb{Z}, +)$ .
- ii. The real numbers form a multiplicative group  $(\mathbb{R}, \times)$ , with  $e = 1$  and  $g^{-1} = \frac{1}{g}, \forall g \in G$ .
- iii. The generalized linear group of degree  $n$ :  $GL_n(\mathbb{F}) = \{A \in \text{Mat}_{n \times n}(\mathbb{F}) : \det(A) \neq 0\}$  is a matrix group under matrix multiplication, with all matrix entries coming from a field  $\mathbb{F}$ . Note in this case  $e = I$ , the identity matrix, and because all matrices are invertible so  $AA^{-1} = A^{-1}A = I, \forall A$ .

#### 1.2 Generators and Relations

Although group elements can be enumerated or expressed by set-builder notation, many groups can be constructed in a more parsimonious and compact way: we introduce the notion of generators and relations.

**Definition 2.** Let  $S$  be a subset of a group  $G$ . We say the “ $S$  generates  $G$ ” if every  $g \in G$  can be written as a finite product of  $s, s^{-1} \in S$ . Notationally, we write  $G = \langle S | R_1, R_2, \dots, R_m \rangle$ , where  $R_1 \dots R_m$  is a collection of relations among  $s \in S$ .

**Example.** (Dihedral Group) Consider a regular  $n$ -gon centered at the origin of the  $x$ - $y$  plane and label the vertices counterclockwise by  $1, 2, 3, \dots, n$ . Let  $r$  be the operation that rotates the  $n$ -gon counterclockwise about the origin by  $2\pi/n$  radian. Let  $s$  be the reflection across the line of symmetry across vertices 1 and  $n$ . Then the Dihedral Group  $D_{2n} = \langle r, s | r^n = e, s^2 = e, rs = sr^{-1} \rangle$ . See Figure 1 for a more concrete example when  $n=3$ .



**Figure 1.** A demonstration of  $D_6$ :  $r, s$  operate on the vertices of an equilateral triangle.

**Remark.** The idea and notation of group generator are of critical importance for the representation of Lie group—a continuous group that help us to solve ODEs. But the generator in Lie group’s case, of course, will not be as straightforward as in this discrete case. The above example of the dihedral group is also purposefully selected because it is a case where the operations preserve the shape of rigid geometric objects, providing a conventional “look-the-same” characterization of symmetries. This paves the way for the following study of how actions carried out by Lie groups leads to the symmetry of ODEs.

### 1.3 Group Action

**Definition 3.** A group  $G$  acting on a finite set  $X$  is:  $G \curvearrowright X: G \times X \rightarrow X$ , defined by left multiplication  $g \cdot x = gx, \forall g \in G$  and  $x \in X$ . Alternatively, we can also think of group action as a permutation of elements in  $X$ :  $G \curvearrowright X: G \rightarrow S_X$ , where  $S_X$  is the set of permutations of  $X$ .

**Definition 4.** The orbit of  $x \in X$  under the group action is  $\mathcal{O}_x = \{g \cdot x: g \in G\}$ .

**Remark.**  $\mathcal{O}_x$  can be understood as “all the places” that  $x$  goes to after the action.

## 2 Ordinary Differential Equations

**Definition 5.** Ordinary Differential Equations are equations that relate a function  $y = y(x)$  to its derivatives  $y, y', y'', \dots, y^{(n)}$ . More formally represented as:

$$y^{(n)} = \omega(x, y, y', y'', \dots, y^{(n-1)}), \text{ for some function } \omega.$$

We call  $x$  independent variable, and  $y$  dependent (or state) variable. The index  $n$  indicates the highest order of derivative.

The goal is often to find a family of solutions, namely a set of  $y$ ’s to the ODE above. Many introductory ODE courses focus on a variety of ad hoc methods that are tailored to some specific types of ODEs, including the method of integrating factor. Those are also referred to as the “standard methods”. However, the rationale behind those methods is more than often not explicated to students.

## 3 Linear Algebra

There is only one concept from Linear Algebra that will become especially handy later in this article.

**Definition 6.** Let  $\{X_1, X_2, \dots, X_R\}$  be a set of linearly independent vectors and  $\mathcal{L} = \text{span}\{X_1, \dots, X_R\}$ , we say  $\{X_1, X_2, \dots, X_R\}$  is a basis for the vector space  $\mathcal{L}$ , and  $\mathcal{L}$  has dimension  $R$ .

## Motivation: unsolvable ODEs?

### Separable ODEs

Recall our definition of an ODE, it is simply an equation that contains a function  $y$ , implicitly defined in  $x$ , and all of its derivatives. Students in an ODE course would be especially pleased if they see the following ODE:

$$y' = \frac{dy}{dx} = \omega(x), \text{ for some generic function } \omega. \quad (1)$$

This type of ODE is considered easily to solve since it involves only the first-order derivative and the derivative only depends on a function of  $x$  and no  $y$  is involved. It belongs to the type called the “separable ODE” [4], whose family of solutions can be found by separating the differentials and integrating on both sides:

$$\begin{aligned} \frac{dy}{dx} &= \omega(x) \\ \int dy &= \int \omega(x) dx \\ y &= \Omega(x) + c, \end{aligned}$$

where  $\Omega$  is the function after integrating and  $c$  is the integrating constant resulting from the indefinite integral. Of course, this same trick can be applied to ODEs of the following forms:

$$\frac{dy}{dx} = g(y) \quad (2)$$

$$\frac{dy}{dx} = g(y) \cdot h(x). \quad (3)$$

### The Integrating Factor Method

In addition to separable ODEs, linear first-order ODEs also are equipped with a well-taught “standard method” known as the integrating factor. Consider the ODE of the form:

$$y' + p(x)y = q(x). \quad (4)$$

The method of integrating factor works as the following: choosing an integrating factor of the form:

$$\mu(x) = \exp\left\{\int p(x)dx\right\}$$

and multiply  $\mu(x)$  through equation 4 to derive a general solution (see [4]):

$$\begin{aligned} \mu(x)(y' + p(x)y) &= \mu(x)q(x) \\ (\mu(x)y)' &= \mu(x)q(x) \\ y(x) &= e^{-\int p(x)dx} \cdot \int e^{\int p(x)dx} q(x)dx + C \cdot e^{-\int p(x)dx}. \end{aligned}$$

## “Unsolvable” ODEs?

Despite the bizzare-looking form, at least the linear first-order ODE can be solved by this systematic method above. However, beginners to ODEs will often consider many other ones as “unsolvable” by standard methods. As an illustration, consider the following example.

**Example 2.** (Riccati Equation)  $\frac{dy}{dx} = xy^2 - \frac{2y}{x} - \frac{1}{x^3}, x \neq 0$ .

This example is neither separable nor solvable by those so-called “standard methods”. It begs the question then, given any ODE, can we find a generalizable, systematic way of finding its family of solutions? The answer is a definite yes for first-order ODEs—I will demonstrate later that every first-order ODE can be reduced to a quadrature (i.e. it is solvable up to an integral). However, the situation will be more nuanced for a generic  $n$ -th order case. The central method we apply here to find solutions is the idea of symmetry, which can turn a seemingly unsolvable ODE into a different coordinate system in which it becomes solvable. The next chapter introduces and fully develops the idea of Lie group of symmetries.

## Symmetries and One-parameter Local Lie Group

### Symmetries of Rigid Geometry

The word “symmetry” is not an alien word to us. Since elementary school, we have been using it to describe shapes that are symmetric in our daily lives (for example, see Figure 2). However, a more rigorous definition for symmetry is needed for us to use it on ODEs. In fact, symmetry is actually a transformation with some special properties.

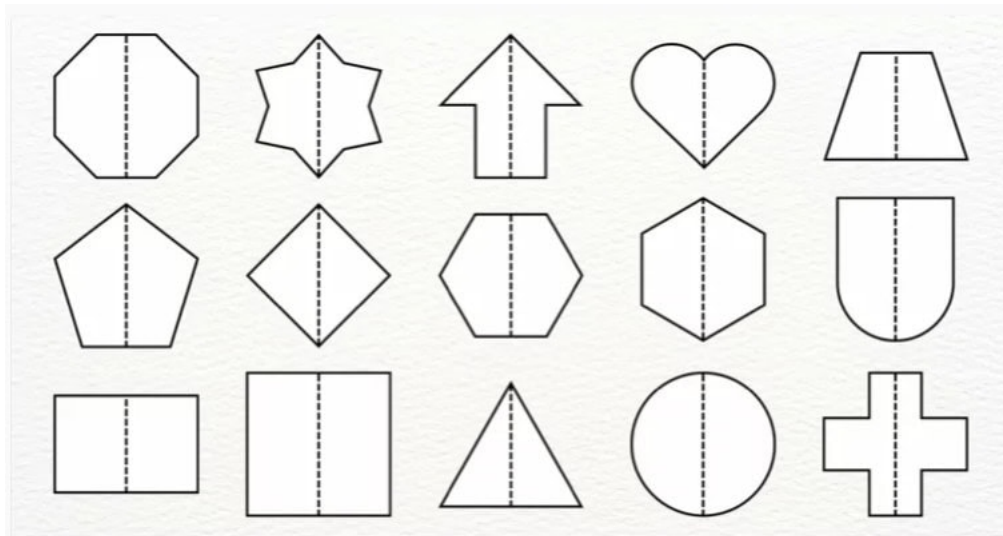


Figure 2.

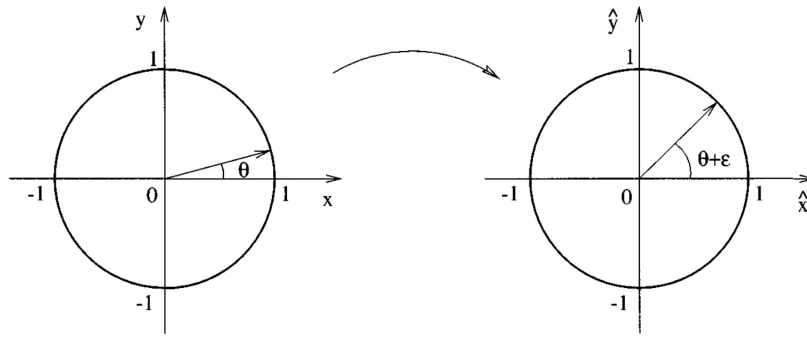
**Definition 7.** A symmetry is a transformation  $\Gamma: x \rightarrow \tilde{x}(x)$ , where  $x$  denotes a generic position of an object, that satisfies the following [3]:

- $\Gamma$  preserves the “structure” of an object.

- $\Gamma$  is a smooth, invertible map whose inverse  $\Gamma^{-1}$  is also smooth.
- $\Gamma$  maps the object to itself.

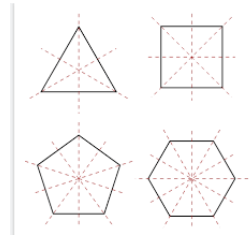
The most visually intuitive case is again the transformation on rigid geometric objects (see Figure 3). Consider the following example involving the unit circle. Let  $\Gamma_\epsilon: (\cos(\theta), \sin(\theta)) \rightarrow (\cos(\theta + \epsilon), \sin(\theta + \epsilon))$ . Note that the unit circle is indistinguishable after rotation;  $\Gamma_\epsilon$  is smooth and has a smooth inverse  $\Gamma_{-\epsilon}$ , which is a rotation by  $-\epsilon$  degree; every point on the circle is mapped to some other point on the circle, so  $\Gamma_\epsilon$  is a map of the circle to itself [3]. Another way of phrasing this symmetry is, of course, resorting to the matrix representation of a rotation of a length-one vector in  $\mathbb{R}^2$ :  $\Psi_\epsilon \cdot \vec{v} = \begin{pmatrix} \cos(\epsilon) & -\sin(\epsilon) \\ \sin(\epsilon) & \cos(\epsilon) \end{pmatrix} \cdot \vec{v}$ .

**Example 3.** (Rigid Rotation of the unit circle is a symmetry)



**Figure 3.** Rotation of the unit circle.

Staying in the realm of rigid geometry for a little while longer, we can find more than just a rotational symmetry. Let's revisit the illustrative example of *Definition 2*. Recall the dihedral group for a rigid, regular n-gon, this group is a set of all symmetries that can be admitted by rigid, regular n-gon. More importantly, rotation and reflection serve as two fundamental building blocks (i.e. generators) to construct other symmetries that do not appear forthcoming (i.e. composition of  $r$ ,  $s$  and their higher orders).



**Figure 4.** Symmetries of rigid, regular n-gons.

## One-parameter Local Lie Group

Going beyond just geometry, we can define a set of transformations more generically in the domain  $\mathbb{R}^2$ . Consider the ordered pairs that denotes positions in  $\mathbb{R}^2$ :  $(x, y)$ , and  $(\tilde{x}, \tilde{y})$ , for some  $\epsilon \in \mathbb{R}$ , let  $\mathcal{P}_\epsilon: (x, y) \rightarrow f(x, y; \epsilon) = (\tilde{x}, \tilde{y})$  be a coordinate transformation depending on *parameter*  $\epsilon$ . Then a set of  $\mathcal{P}_\epsilon$ , call it  $G = \{\mathcal{P}_\epsilon\}_{\epsilon \in \mathbb{R}}$ , is a *transformation group* if the following conditions are met:

**Definition 8.** (*Conditions for a Transformation Group*)

- i.  $\mathcal{P}_\epsilon$  is one-to-one and onto;
- ii.  $\mathcal{P}_{\epsilon_1} \circ \mathcal{P}_{\epsilon_2} = \mathcal{P}_{\epsilon_1 + \epsilon_2}$ ;
- iii.  $\mathcal{P}_{\epsilon=0} = I$ ;
- iv. Every  $\mathcal{P}_\epsilon$  has a unique inverse:  $\forall \mathcal{P}_\epsilon$ , there exists  $-\epsilon$  such that  $\mathcal{P}_\epsilon \circ \mathcal{P}_{-\epsilon} = I$ .

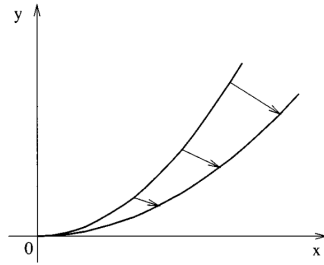
Although  $G$  is a group with respect of functional composition, condition *ii* effectively turns functional composition into real number addition: a sequential application of two transformations with different parameters is equivalent to application of one with the two parameters added. Condition *iii* guarantees the existence of an identity transformation that happens exactly when we set the parameter  $\epsilon = 0$ . Lastly, condition *iv* ensures we can always transform back to the identity status of an object by taking the additive inverse of any given  $\epsilon$ . These three conditions collective endow the set  $G$  group properties [6].

Further more, if the function  $f$  that governs the transformation has two more desirable properties, the group  $G$  becomes a one-parameter local Lie group, or Lie group of point symmetries [6].

**Definition 9.** *The group  $G$  above is called an “one-parameter local Lie group” (or Lie group of point symmetry) if the transformation function  $f$  is infinitely differentiable with respect to  $x$  and analytic with respect to  $\epsilon$ .*

**Note.** We say  $f$  is analytic to  $\epsilon$  if  $f$  can be locally given by a convergent power series centered at some  $\epsilon$  (such as Taylor Series).

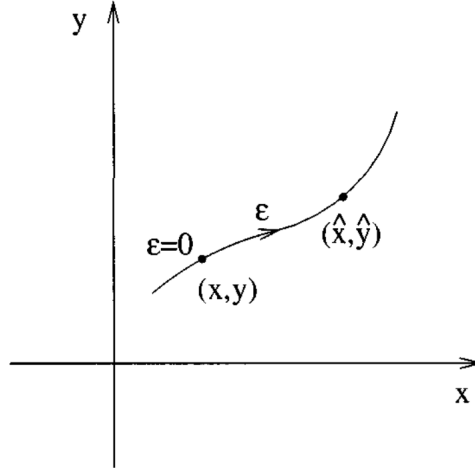
How does the Lie group of point symmetry impact smooth curves in  $\mathbb{R}^2$ , which are usually the solutions to ODEs? Since each group element is a coordinate transformation, the action of a Lie group is unpacked as the following: a solution curve  $y = f(x)$  for a first-order ODE  $y' = \omega(x, y)$  will be mapped to another solution curve  $y = \hat{f}(x)$ :  $(x, f(x)) \rightarrow (\tilde{x}, \hat{f}(x))$ . For instance, Figure 5 visualizes a scaling symmetry given by  $(x, y) = (e^\epsilon x, e^{-\epsilon} y)$ , in which the y-direction experiences an  $\epsilon$ -rate exponential decay while the x-direction an exponential growth.



**Figure 5.** Action of a scaling symmetry on a solution curve.

## Lie Group Action and Orbit

For readers with background in abstract algebra, what a Lie group does to a set of solution curves of a first-order ODE can be framed as a group action  $G \curvearrowright \mathbb{R}^2: \mathbb{R} \times \mathbb{R}^2 \rightarrow \mathbb{R}^2$ , carried out by the mapping:  $((x, y), \epsilon) \rightarrow (\tilde{x}(x, y; \epsilon), \tilde{y}(x, y; \epsilon))$ . Thus, we can study the orbit of this group action, which is a continuous, smooth path traced out in  $\mathbb{R}^2$  by varying the value of  $\epsilon$  [7].



**Figure 6.** A one-dimensional orbit[3].

A helpful heuristic tool to understand the orbit and the role of  $\epsilon$  is to regard the orbit path as a “parametrized” vector function by a parameter  $\epsilon$ , because we can find a vector-valued function  $\vec{r}(t) = \langle x(t), y(t) \rangle$  that parametrizes a vector field  $\vec{F}(x, y) = \langle P(x, y), Q(x, y) \rangle$  [8]. In this article’s setting, the parameter  $\epsilon$  plays the role of  $t$ , but it does not necessarily obtain a time-based interpretation as in multivariable calculus. The orbit of Lie group action is important not only because of its aesthetically pleasing visuals, but also because it provides the information we need to find the generator of the entire Lie group of point symmetry. To see why, recall that we previously required the transformation function to be analytic with respect to  $\epsilon$ , so we can take the first-order Taylor series expansion centered at  $\epsilon = 0$  component-wise:

$$\tilde{x} = x + \epsilon \frac{d\tilde{x}}{d\epsilon} \Big|_{\epsilon=0} + \dots (\text{higher order terms}) \text{ and} \quad (5)$$

$$\tilde{y} = y + \epsilon \frac{d\tilde{y}}{d\epsilon} \Big|_{\epsilon=0} + \dots (\text{higher order terms}). \quad (6)$$

Ignoring the higher-order terms, we notice that the first-order terms provides the tangency and a linearized approximation at each local point on the orbit. The next subsection further develops this idea and demonstrate how knowing the local tangency is sufficient to generate the global transformation.

## The Infinitesimal Generator of Lie Group

Along the orbit of Lie group action, at every point  $(x, y)$  we can find the tangent vector  $\vec{T}(x, y) = \langle \xi(x, y), \eta(x, y) \rangle$ , which is also a smooth vector field that characterizes the “velocity” of the group action[3]. To be more precise, we find the tangent vector at each point  $(x, y)$  by:

$$\xi(x, y) = \frac{d\tilde{x}}{d\epsilon} \Big|_{\epsilon=0}; \eta(x, y) = \frac{d\tilde{y}}{d\epsilon} \Big|_{\epsilon=0}. \quad (7)$$

Notice that the  $(x, y)$  is the “starting point” of transformation, so  $(\tilde{x}, \tilde{y}; \epsilon = 0) = (x, y)$ . We can also find the tangency at each  $(\tilde{x}, \tilde{y})$  further along the orbit too by setting:

$$\xi(\tilde{x}, \tilde{y}) = \frac{d\tilde{x}}{d\epsilon}; \eta(\tilde{x}, \tilde{y}) = \frac{d\tilde{y}}{d\epsilon}.$$

We further formally define the infinite generator as the following [3].

**Definition 10.** Let  $G$  be a one-parameter local Lie group, it has generator of the form :

$$X = \xi(x, y) \frac{\partial}{\partial x} + \eta(x, y) \frac{\partial}{\partial y} \quad (8)$$

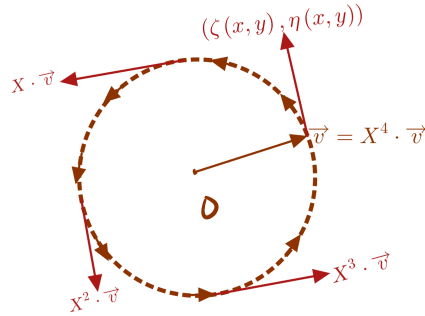
where  $\xi(x, y) = \frac{d\tilde{x}}{d\epsilon}|_{\epsilon=0}$  and  $\eta(x, y) = \frac{d\tilde{y}}{d\epsilon}|_{\epsilon=0}$ , representing the horizontal and vertical component of the tangent vector at  $(x, y)$ , respectively.

**Remark.** The generator  $X$  is an "operator". For instance,  $X \cdot h = \xi(x, y) \frac{\partial h}{\partial x} + \eta(x, y) \frac{\partial h}{\partial y}$ . It takes in a function and spits out another.

**Notation.** Thereafter, I will suppress the arguments and use  $\xi = \xi(x, y)$  and  $\eta = \eta(x, y)$  and use  $\partial x$  and  $\partial y$  (or simply right subscript of  $x$  and  $y$ ) in place of the partial derivative operator  $\frac{\partial}{\partial x}$ , and  $\frac{\partial}{\partial y}$ , for notational simplicity.

The general idea is that repeated application of local tangent vectors can generate the entire global transformation. Although it is hard to visualize this generator in a generic ODE's case, we can further build on the case of rotation in  $\mathbb{R}^2$  by revisiting Example 3•. Consider the action  $\mathbb{R} \times \mathbb{R}^2 \rightarrow \mathbb{R}^2: (\epsilon, \vec{v}) \rightarrow \Psi_\epsilon \cdot \vec{v}$ , where  $\Psi_\epsilon = \begin{pmatrix} \cos(\epsilon) & -\sin(\epsilon) \\ \sin(\epsilon) & \cos(\epsilon) \end{pmatrix}$ . Differentiating  $\Psi_\epsilon$  and setting  $\epsilon = 0$ , we have:

$$\frac{d}{d\epsilon} \Psi_\epsilon|_{\epsilon=0} = \begin{pmatrix} -\sin(0) & -\cos(0) \\ \cos(0) & -\sin(0) \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = X \quad (9)$$



**Figure 7.** A visualization of how  $X = \frac{d}{d\epsilon} \Psi_\epsilon|_{\epsilon=0}$  generates a  $2\pi$  rotation (Author's own note).

Observe that the generator  $X$  is an element of the rotation group  $\text{SO}(2) = \{A \in \text{Mat}_{2 \times 2}(F) : \det(A) = 1\}$ , representing a  $\frac{\pi}{2}$  counterclockwise rotation of any vector at the starting point  $\vec{v} = \begin{pmatrix} x \\ y \end{pmatrix}$ . As demonstrated in Figure 7, applying the operator  $X$  to vector  $\vec{v}$  four times will complete a  $2\pi$  rotation that returns to the starting position  $\vec{v} = \begin{pmatrix} x \\ y \end{pmatrix}$ . In a more generic case where the tangent vector represents a less visualizable vector field, we can conceptualize its generator property by repeatedly integrating over the flow to generate the group orbit's pathlines. Now we have developed all the theoretical tools regarding symmetries, the next chapter applies them to a focused study of first-order ODEs.

# How to Use Symmetry to Solve (any) First-order ODEs

## The Symmetry Condition

Given a Lie group of symmetry generated by  $X = \xi \partial x + \eta \partial y$  and a first-order ODE of the form:

$$\frac{dy}{dx} = \omega(x, y), \quad (10)$$

we say ODE 10 admits this Lie group of symmetry if:

$$\frac{d\tilde{y}}{d\tilde{x}} = \omega(\tilde{x}, \tilde{y}) \text{ when } \frac{dy}{dx} = \omega(x, y). \quad (11)$$

Condition 11 is known as the *Symmetry Condition*, which is a manifestation of the central idea of symmetry—the image ODE looks the same post-transformation. The symmetry condition is captured more precisely by the following definition, through conducting implicit differentiation on the left-hand side of the equation.

**Definition 11.** (*Symmetry Condition*) For a first-order ODE  $y' = \omega(x, y)$ , it admits a Lie symmetry  $\mathcal{P}_\epsilon: (x, y) \rightarrow (\tilde{x}, \tilde{y}; \epsilon)$  if:

$$\frac{d\tilde{y}}{d\tilde{x}} = \omega(\tilde{x}, \tilde{y}).$$

Expanding the right side with implicit differentiation, we have:

$$\frac{d\tilde{y}}{d\tilde{x}} = \frac{\frac{d}{dx}(\tilde{y})}{\frac{d}{dx}(\tilde{x})} = \frac{\tilde{y}_x + y' \tilde{y}_y}{\tilde{x}_x + y' \tilde{x}_y} = \frac{\tilde{y}_x + \omega \cdot \tilde{y}_y}{\tilde{x}_x + \omega \cdot \tilde{x}_y}, \quad (12)$$

where  $d/dx$  is the “total derivative operator” [3].

Below is a worked-out example that helps the readers to internalize Condition 11 and 12.

**Example 4.** Consider the previous scaling symmetry in Figure 5:  $(x, y) \rightarrow (e^\epsilon x, e^{-\epsilon} y)$ . We show that the ODE  $y' = \frac{2y}{x}$  admits this scaling symmetry.

$$\begin{aligned} \frac{d\tilde{y}}{d\tilde{x}} &= \frac{\tilde{y}_x + \omega \cdot \tilde{y}_y}{\tilde{x}_x + \omega \cdot \tilde{x}_y} = \frac{0 + \frac{2y}{x} e^{-\epsilon}}{e^\epsilon + 0} \\ &= \frac{2(y e^{-\epsilon})}{x e^\epsilon} = \frac{2\tilde{y}}{\tilde{x}}, \text{ so we conclude that:} \\ \frac{d\tilde{y}}{d\tilde{x}} &= \omega(\tilde{x}, \tilde{y}) \end{aligned}$$

## Reconstructing Lie Symmetries from the Infinitesimal Generator

It would be fruitful to guess and check what symmetries a first-order ODE might admit. As Starrett [6] suggests, trying Lie symmetries with  $\xi = P(x), \eta = Q(y)$ ;  $\xi = Q(y), \eta = P(x)$ ;  $\xi = P_1(x) + Q_1(y), \eta = P_2(x) + Q_2(y)$ ; their products; quotients; etc. However, there is a more systematic way of reconstructing Lie symmetries just by leveraging the symmetry condition 12 and the coordinate-wise Taylor series linearization 6 and 5 of the orbit pathline. Notice linearization 6 and 5 can be rewritten with the symble  $\xi$  and  $\eta$ :

$$\begin{aligned} \tilde{x} &= x + \epsilon \xi + \dots \\ \tilde{y} &= y + \epsilon \eta + \dots \end{aligned}$$

Putting together the two equations above, we can linearize the image of transformed ODE by:

$$\omega(\tilde{x}, \tilde{y}) = \omega(x, y) + \epsilon(\omega_x \xi + \omega_y \eta), \text{ ignoring higher-order terms.} \quad (13)$$

But also notice that by working with the linearized orbit we obtain:

$$\omega(\tilde{x}, \tilde{y}) = \frac{\epsilon \eta_x + \omega(x, y) + \epsilon \omega(x, y) \eta_y}{1 + \epsilon \xi_x + \epsilon \omega(x, y) \xi_y} = \frac{\omega + \epsilon(\eta_x + \omega \eta_y)}{1 + \epsilon(\xi_x + \omega \xi_y)}. \quad (14)$$

By equating Equation 13 and 14, we further simplify to derive the *linearized symmetry condition*:

$$\eta_x - \xi_y \omega^2 + (\eta_y - \xi_x) \omega - (\xi \omega_x + \eta \omega_y) = 0. \quad (15)$$

In practice, we expand Equation 15 by plugging in the known information, collect terms, and then set coefficients in front of powers of  $y$  to zero. This computation will usually generate a system of partial differential equations (PDEs), but each of them is typically easier to solve than the original ODE. By solving those PDEs, we may be able to find determining information regarding  $\xi$  and  $\eta$ , thus reverse-engineer to integrate to retrieve  $\tilde{x}(x, y; \epsilon)$  and  $\tilde{y}(x, y; \epsilon)$ .

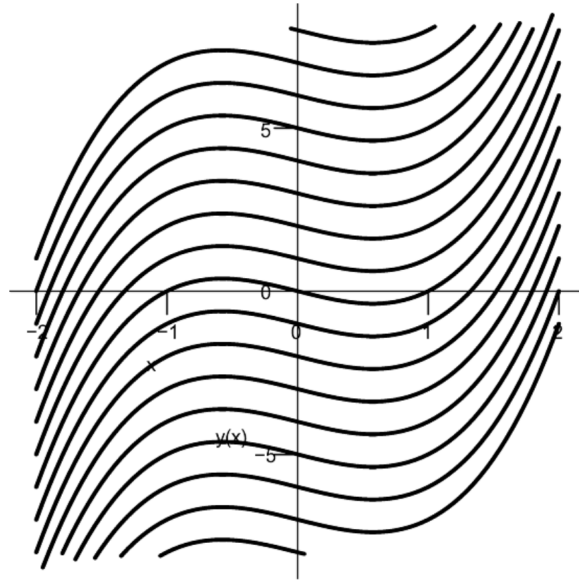
## Solving Any First-order ODEs

### Separable ODEs Revisited

Recall one of the simplest ODE:  $y' = f(x)$ , which can be easily solved by separation of variable and then by integration techniques  $y(x) = \int f(x) dx + \alpha = \phi(x; \alpha)$ , where  $\alpha$  is the constant of integration. In fact, we can do this exactly because of a Lie symmetry at work: a translation by  $\alpha$  only in the  $y$ -direction:

$$(x, y) \rightarrow (x, y + \alpha). \quad (16)$$

The solution curves to  $y' = f(x)$  under symmetry 16 simply “slides into” one another under the action of this  $y$ -translation symmetry[6].

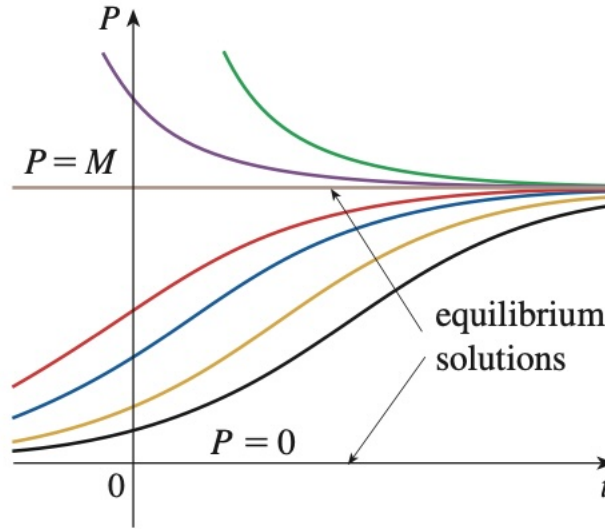


**Figure 8.** The slope of  $y(x)$  only depends on  $x$  [6].

A similar argument can be made for the case where the ODE is only dependent on a function of  $y$ . The ODE of the form  $y' = g(y)$  has solution curves  $y = \psi(y; \beta)$ , which corresponds to a translation only in the  $x$ -direction:

$$(x, y) \rightarrow (x + \beta, y). \quad (17)$$

As Figure 9 illustrates, solutions curves of ODEs under symmetry 17 looks indistinguishable up to a translation in the  $x$ -direction.



**Figure 9.** Solution Curves  $P(t)$  to Logistic Equation  $\frac{dP}{dt} = kP(1 - \frac{P}{M})$ , where  $P$  is the population as a function of time,  $k$  is the rate of growth and  $M$  is the “carrying capacity” of a habitat.

## Canonical Coordinate

Of course, not every ODE can be solved by method of separation, since the symmetry condition 11 may collapse for so many ODEs not admitting a  $y$ -translation symmetry. But what we can do instead is to manually construct a new coordinate system in which the ODE does admit a uni-direction translational symmetry.

Provided that a first-order ODE admits a known Lie point symmetry generated by  $X = \xi \partial_x + \eta \partial_y$ , we make the change of coordinate from  $(x, y)$  to  $(r, s)$  such that the ODE in the second coordinate system becomes separable, or equivalently, it admits the  $s$ -translation symmetry  $(r, s) \rightarrow (r, s + \epsilon)$ . Then the first-order ODE 10 will be reduced to one that's only dependent on a function containing  $r$ :

$$\frac{dy}{dx} = \omega(x, y) \Rightarrow \frac{ds}{dr} = \Omega(r) \quad (18)$$

, where  $(r, s) = (r(x, y), s(x, y))$  is called the *Canonical Coordinate*. Also immediately notice the tangent vector at any point  $(r, s)$  now becomes:  $\left\langle \frac{dr}{d\epsilon} \Big|_{\epsilon=0}, \frac{ds}{d\epsilon} \Big|_{\epsilon=0} \right\rangle = \langle 0, 1 \rangle$ . A closer look at the influence of coordinate change on the generator  $X$  can help us compute the explicit form of  $r(x, y)$ ,  $s(x, y)$ .

We firstly show how the operator  $X$  will change under a generic coordinate change  $(x, y) \rightarrow (u(x, y), v(x, y))$ . Let  $F(u, v)$  be a smooth function and we apply  $X$  on  $F$  to obtain:

$$\begin{aligned}
X \cdot F &= (\xi \partial x + \eta \partial y) \cdot F \\
&= \xi \frac{\partial F(u, v)}{\partial x} + \eta \frac{\partial F(u, v)}{\partial y} \\
&= \xi \left( \frac{\partial F}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial F}{\partial v} \frac{\partial v}{\partial x} \right) + \eta \left( \frac{\partial F}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial F}{\partial v} \frac{\partial v}{\partial y} \right) \text{ (by chain rule)} \\
&= \frac{\partial F}{\partial u} (\xi u_x + \eta u_y) + \frac{\partial F}{\partial v} (\xi v_x + \eta v_y) \\
&= F_u X \cdot u + F_v X \cdot v.
\end{aligned} \tag{19}$$

So the generator under the coordinate  $(u, v)$  becomes:

$$X = X \cdot u \frac{\partial}{\partial u} + X \cdot v \frac{\partial}{\partial v}. \tag{20}$$

Now apply result 20 to canonical coordinates  $(r, s)$ , we have:

$$\begin{aligned}
X &= X \cdot r \frac{\partial}{\partial r} + X \cdot s \frac{\partial}{\partial s} \\
&= \frac{\partial}{\partial s}, \text{ since } X \cdot r = 0 \text{ and } X \cdot s = 1.
\end{aligned} \tag{21}$$

It can be concluded that the generator under the canonical coordinate takes a rather simplified form:  $X = \partial s$ . Moreover, based on the observation that the tangent vector is  $\langle 0, 1 \rangle$  at any  $(r, s)$ , we use chain rule to further expand and obtain:

$$\frac{d\tilde{r}}{d\epsilon}|_{\epsilon=0} = \frac{\partial \tilde{r}}{\partial x} \frac{dx}{d\epsilon}|_{\epsilon=0} + \frac{\partial \tilde{r}}{\partial y} \frac{dy}{d\epsilon}|_{\epsilon=0} = \frac{\partial r}{\partial x} \xi + \frac{\partial r}{\partial y} \eta = 0, \tag{22}$$

$$\frac{d\tilde{s}}{d\epsilon}|_{\epsilon=0} = \frac{\partial \tilde{s}}{\partial x} \frac{dx}{d\epsilon}|_{\epsilon=0} + \frac{\partial \tilde{s}}{\partial y} \frac{dy}{d\epsilon}|_{\epsilon=0} = \frac{\partial s}{\partial x} \xi + \frac{\partial s}{\partial y} \eta = 1. \tag{23}$$

Equation 22 can be rewritten as  $\langle \xi, \eta \rangle \cdot \vec{\nabla} r = 0$ , meaning the surface defined by  $r(x, y) = c$ , for some constant  $c$  along the direction pointed in the direction  $\langle \xi, \eta \rangle$ . By the method of characteristics, we say the slope  $\frac{dy}{dx} = \frac{\eta}{\xi}$  and the constant  $c$  is the integration constant of computing:

$$\int \frac{dx}{\xi} = \int \frac{dy}{\eta}. \tag{24}$$

Similarly, we rewrite Equation 23 into  $\langle \xi, \eta, -1 \rangle \cdot \left\langle \frac{ds}{dx}, \frac{ds}{dy}, 1 \right\rangle = 0$ . Again, by the method of characteristics, we obtain  $\frac{dy}{\eta} = \frac{dx}{\xi} = ds$ . Hence, we can find the closed form of  $s(x, y)$  through integration:

$$\int ds = s = \int \frac{dx}{\xi} = \int \frac{dy}{\eta}. \tag{25}$$

Now that we have successfully find the explicit form  $(r(x, y), s(x, y))$ , we substitute them into the original ODE 10 by leveraging the symmetry condition and the  $s$ -translation symmetry:

$$\frac{ds}{dr} = \frac{\frac{d}{dx}(s)}{\frac{d}{dx}(r)} = \frac{\frac{\partial s}{\partial x} + \omega \frac{\partial s}{\partial y}}{\frac{\partial r}{\partial x} + \omega \frac{\partial r}{\partial y}} = \Omega(r). \tag{26}$$

From this point forward, one can comfortably apply separation of variable on ODE 26:

$$\begin{aligned}\int ds &= \int \Omega(r) dr + C \\ s &= \int \Omega(r) dr + C.\end{aligned}$$

To be complete, we can always back-substitute  $(x, y)$  to find solutions as a function with only variables  $x, y$ :

$$s(x, y) = \int^{r(x, y)} \Omega(r) dr + C. \quad (27)$$

## A Road Map

So far, we have demonstrated the powerfulness of Lie's method in solving first-order ODEs. By leveraging the symmetry condition and canonical coordinate to reduce the ODE to an integral like Equation 27. Below I provide a roadmap to a systematic method to craft solutions to any first-order ODEs.

Given  $y' = \omega(x, y)$  and one of its known Lie symmetries generated by  $X = \xi \partial x + \eta \partial y$ :

- Leverage the generator  $X$  under canonical coordinate;
- Find expressions for  $(r, s)$  by method of characteristics;
- Substitute  $(r, s)$  into  $y' = \omega(x, y)$  to obtain  $\frac{ds}{dr} = \Omega(r)$ ;
- Integrate  $s = \int \Omega(r) dr + C$ ;
- Back-substitute to get  $s(x, y) = \int^{r(x, y)} \Omega(r) dr + C$ , a general solution in  $x$  and  $y$ .

In terms of generalizability, this method is guaranteed to work for all first-order ODEs because of the theorem below.

**Theorem 1.** *Every first-order ODE admits infinitely many groups of symmetry.*

## Lie Symmetry and $n$ -th Order ODEs

The previous chapter has demonstrated the powerfulness of Lie's method in solving any first-order ODEs. But what about second, third, or a generic  $n$ -th order ODE? Can Lie's method still be applied? If so, does it have the same level of power as in the first-order case? This section extends the tools of symmetry developed in previous sections to the generic  $n$ -th order ODEs. We are now concerned with ODE of the form:

$$y^{(n)} = \omega(x, y, y', y'', \dots, y^{(n-1)}), \text{ where } y^{(k)} = \frac{d^k y}{dx^k}, \text{ for } k = 1, 2, 3, \dots, n. \quad (28)$$

We will first discuss the symmetry condition, linearized symmetry condition, and prolonged infinitesimal generator for ODE 28. Then we introduce the concept of  $R$ -parameter Lie group. Lastly, we briefly summarize how the  $R$ -parameter Lie group can help us reduce the order of a higher-order ODE to some degree, although no satisfying recipe like the one designed for first-order ODEs exists.

## Symmetry Condition

Intuitively, the action of Lie group on the domain  $\mathbb{R}^2$ :  $(x, y) \rightarrow (\tilde{x}, \tilde{y})$  also induces an action on all derivatives  $y^{(k)}$ , for  $k = 1, 2, 3, \dots, n$  by a mapping on an extended variable space:

$$\Gamma: (x, y, y', y'', \dots, y^{(n)}) \rightarrow (\tilde{x}, \tilde{y}, \tilde{y}', \tilde{y}'', \dots, \tilde{y}^{(n)}), \quad (29)$$

where  $\tilde{y}^{(k)}$  is iteratively defined for each  $k$  as:

$$\tilde{y}^{(k)} = \frac{d\tilde{y}^{(k-1)}}{d\tilde{x}} = \frac{\frac{d}{dx}(\tilde{y}^{(k-1)})}{\frac{d}{dx}(\tilde{x})} [3]. \quad (30)$$

In this case the total derivative operator  $d/dx = \partial x + y' \partial y + y'' \partial y' + \dots + y^{(n)} \partial y^{(n-1)}$ . With the extension of this action to the higher-order derivatives, we then formally define the symmetry condition for an  $n$ -th order ODE as the following.

**Definition 12.**  $\tilde{y}^{(n)} = \omega(\tilde{x}, \tilde{y}, \tilde{y}', \tilde{y}'', \dots, \tilde{y}^{(n-1)})$  when  $y^{(n)} = \omega(x, y, y', y'', \dots, y^{(n-1)})$ .

## Linearized Symmetry Condition

Since component-wise the transformation that governs the mapping described by Equation 29 is infinitely differentiable and analytic with respect to  $\epsilon$ , we can safely apply the first-order Taylor series expansion centered at  $\epsilon = 0$ :

$$\begin{aligned} \tilde{x} &= x + \epsilon \left. \frac{d\tilde{x}}{d\epsilon} \right|_{\epsilon=0} + \dots = x + \epsilon \xi + \dots \\ \tilde{y} &= y + \epsilon \left. \frac{d\tilde{y}}{d\epsilon} \right|_{\epsilon=0} + \dots = x + \epsilon \eta + \dots \\ \tilde{y}' &= y' + \epsilon \left. \frac{d\tilde{y}'}{d\epsilon} \right|_{\epsilon=0} + \dots = x + \epsilon^{(1)} \eta + \dots \\ &\vdots \\ \tilde{y}^{(n)} &= y^{(n)} + \epsilon \left. \frac{d\tilde{y}^{(n)}}{d\epsilon} \right|_{\epsilon=0} + \dots = x + \epsilon^{(n)} \eta + \dots, \end{aligned}$$

where

$$^{(s)}\eta = \left. \frac{d\tilde{y}^{(s)}}{d\epsilon} \right|_{\epsilon=0}, \text{ for } s = 1, 2, \dots, n, \text{ and } ^{(0)}\eta = \eta [7].$$

## Prolonged Infinitesimal Generator

The linearized symmetry condition above immediately motivates us to put together a tangent vector corresponding to the variable space  $(x, y, y', y'', \dots, y^{(n)})$  as  $\vec{T} = \langle \xi, \eta, ^{(1)}\eta, \dots, ^{(n)}\eta \rangle$ , which immediately motivates the definition of the *prolonged infinitesimal generator* [3].

**Definition 13.** The  $n$ -th prolongation of generator  $X = \xi \partial x + \eta \partial y$  is

$$X^{(n)} = \xi \partial x + \eta \partial y + ^{(1)}\eta \partial y' + \dots + ^{(n)}\eta. \quad (31)$$

The prolonged generator can help us write the symmetry condition in Definition 12 in a more compact notation. It turns out that Definition 12 is equivalent to having the generator  $X^{(n)}$  on both sides of Equation 28:

$$\begin{aligned} X^{(n)} \cdot y^{(n)} &= X^{(n)} \cdot \omega(x, y, y', y'', \dots, y^{(n-1)}) \\ ^{(n)}\eta &= X^{(n)} \cdot w [7]. \end{aligned}$$

## ***R*-parameter Local Lie Group**

Although the tool of symmetry extends perfectly well into the variable space  $(x, y, y', y'', \dots, y^{(n)})$ , it does not provide an all-powerful recipe to find a solution to the  $n$ -th order ODEs. The gist is that higher-order  $n$  places restrictions on the number of Lie symmetries we can find, denoted as  $R$  [3]. In another word, we are no longer able to rely on the existence infinitely many Lie groups of symmetries as in the first-order case. Nonetheless, a finite set of Lie symmetries can still be of avail in simplifying a  $n$ th-order ODE. Before we unpack when and how the reduction of order could be done, another new terminology has to be introduced.

Given an ODE of the form of Equation 28 with  $n \geq 2$  and  $R$  one-parameter local Lie groups it admits, we have a collection of generator  $\{X_1, X_2, \dots, X_R\}$  in which each one generates its corresponding Lie group of point symmetry. To be more precise, each  $X_i = \xi_i \partial x + \eta_i \partial y$ . We then denote  $\mathcal{L}^{(R)} = \text{span}\{X_1, X_2, \dots, X_R\}$  as a vector space. We say the set  $\{X_1, X_2, \dots, X_R\}$  is a basis for  $\mathcal{L}^{(R)}$ . Any other (generator of) Lie symmetry admitted by this ODE can then be written as a linear combination of all the basis elements (see Equation 32) [3].

$$X = \sum_{i=1}^R c_i X_i \quad (32)$$

Then, we say the group generated by the spanning set  $\mathcal{L}^{(R)}$  is a *R-parameter local Lie group*.

## **Reduction of order and R-parameter Lie Group**

As suggested above, the higher-order  $n$  induces many restrictions on  $R$ . For instance, for second-order ODEs, the number  $R$  falls in  $\{0, 1, 2, 3, 8\}$ . Stephani's text [7] offers an elegant proof by contradiction for the theorem that *for a second-order ODE, the maximum of symmetries it can admit is eight*. Stephani [7] also furnishes an second-order ODE where no symmetries exist! Readers are encouraged to consider the ODE  $y'' = xy + e^{y'} + e^{-y'}$  and verify  $R = 0$  in this case [7]. Moreover, for any ODE with  $n \geq 3$ , it has  $R \leq n + 4$ . Replicating the proofs of those theorems is beyond the scope of this expository work, but we consider a simplified case in which if we find one symmetry, we can achieve a reduction of order by one.

### **Reduction of Order to $(n - 1)$**

In a more application-oriented setting, researchers who use ODEs to model natural or scientific phenomenon are concerned with ODEs with highest order of two or three. For instance, in the formula that articulates Newton's Second Law of Motion  $\sum F = m \frac{d^2x}{dt^2}$ , the second-order derivative  $\frac{d^2x}{dt^2} = a(t)$  is the acceleration of a moving object, and  $x(t)$  denotes its position as a function of time. Physicists refer to  $\frac{d^3x}{dt^3} = a'(t)$  as the "jerk", which measures an abrupt change in the rate of acceleration, but derivatives of fourth, fifth, and higher orders in this case usually lose substantive interpretability<sup>1</sup>. The core message is that since on most occasions we deal with ODEs of order two and three, it would be helpful enough if we can simply reduce the order by one.

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1. Admittedly, physicists do have corresponding terms for the fourth, fifth, and sixth order of derivatives of the position function: snap, crack, and pop, respectively.

Consider the following ODE:

$$\omega(x, y, y', \dots, y^{(n)}) = 0, \quad (33)$$

which admits a Lie group of symmetry with infinitesimal generator  $X = \xi \partial x + \eta \partial y$ .

We again leverage the canonical coordinate  $(r, s)$ . We know ODE 33 reduces to:

$$\Omega(r, s, s', s'', \dots, s^{(n)}) = 0. \quad (34)$$

However, since ODE 33 is invariant under the the s-direction translation, we have

$$\frac{\partial \Omega}{\partial s} = 0 \text{ [3].}$$

So  $\Omega$  does not contain the variable  $s$ , which turns ODE 34 into

$$\Omega(r, s', s'', \dots, s^{(n)}) = 0.$$

By setting a new variable  $z = \frac{ds}{dr} = s'$ , we obtain

$$\Omega(r, z, z', \dots, z^{(n-1)}) = 0 \quad (35)$$

, which is a reduction of order by one [1].

### Reduction of Order to $(n - R)$

Numerous constraints from  $n$  and  $R$  notwithstanding, under some conditions it is theoretical possible to achieve a  $(n - R)$  reduction of order. Below I present this theorem and offer some comments.

**Theorem.** *Assume an  $R$ -parameter Lie group ( $R \geq 3$ ) is admitted by an  $n$ th-order ODE ( $n \geq R$ ). We can have a reduction to an  $(n - R)$ th-order ODE plus  $R$  quadratures if and only if the  $\mathcal{L}^{(R)}$  is a solvable Lie algebra.*

For audience interested in the abstract algebra background behind this theorem, it would be helpful to know that  $\mathcal{L}^{(R)}$  is a solvable Lie algebra *if and only if* there exists a chain of subalgebras

$$\mathcal{L}^{(1)} \subset \mathcal{L}^{(2)} \subset \mathcal{L}^{(3)} \subset \dots \subset \mathcal{L}^{(R)}$$

where  $\mathcal{L}^{(k)}$  is a  $k$ -dimensional Lie algebra and  $\mathcal{L}^{(k-1)}$  is an ideal of  $\mathcal{L}^{(k)}$ , for  $k = 1, 2, \dots, R$  [1]. In practice, by choosing a canonical coordinate  $(r, s)$ , we can make a further coordinate change by setting a new variable  $v = \frac{ds}{dr}$  to reduce ODE into:

$$v_R^{(n-R)} = \Omega(r_R, v_R, \dots, v_R^{(n-R-1)}); v_R^{(k)} = \frac{d^{(k)}v_R}{dr_R^{(k)}}, \text{ for } k = 1, 2, \dots, n - R \quad (36)$$

, which are an  $(n - R)$ -order ODE plus  $R$  quadratures [3]. The actual process to achieve result 36 is too computationally cumbersome to present in this writing, given the space limit; different authors tend to use slightly different approaches. Hydon [3] demonstrates this through solving a matrix system by performing Gauss-Jordan elimination, while Bluman's work [1] entertains a more algorithmic flavor and utilizes the commutator relations among generators  $X_1, X_2, \dots, X_R$ , which could be more appealing to readers with background knowledge of Lie algebra and Lie bracket.

## Conclusion and Discussion

So far, this paper has explored a handful of results in order to unravel the powerfulness of the theory of Lie group in crafting solutions to ODEs. We first motivated the readers to reflect upon the “solvability” of ODEs and the limitation of those so-called “standard methods” that are commonly taught in ODE courses. Then we introduced the concept of symmetry, formulated the algebraic tool of one-parameter local Lie group, including tangent vectors and the infinitesimal generator. With the help of Lie groups, we then applied the tool to first-order ODEs: we defined the symmetry condition, demonstrated how to reconstruct Lie symmetries, and built canonical coordinate in which all first-order ODEs with a known Lie symmetry can be reduced to a problem of quadrature. In the second half of this paper, we extend into the generic  $n$ -th order case. Although the tool of Lie symmetry extends intuitively into the variable space  $(x, y, y', y'', \dots, y^{(n)})$ , including the symmetry conditions and prolonged infinitesimal generator, Lie’s method is limited by the higher order that places restrictions on the number of Lie symmetries. At the end, we demonstrated in the higher-order case, a  $R$ -parameter Lie group can bring about a reduction of order under some scenarios. For an application purpose, it is helpful to simply reduce the order  $n$  to  $(n - 1)$ , while more generically we could achieve a  $(n - R)$  reduction of order when we have a solvable Lie algebra.

It is worthy to point out that although Lie’s theory appears more often in other modern-day fields of mathematics, especially in quantum physics, Sophus Lie’s original source of inspiration was indeed rooted in solving differential equations [5]. What he discovered was an ODE-analog to the *Galois Theory*, which addresses the solvability of a degree- $n$  polynomial. Suggestive in notation  $n$ , it perhaps strikes an immediate similarity with the solvability of a  $n$ -th order ODEs discussed in this article. It turns out that Lie group is the manifestation of the spirit of Galois group in the realm of ODEs. On one hand, the Galois group is a symmetry group that contains all permutations of roots of a polynomial that preserves its solvability (i.e. the roots can be solved by radicals). On the other, the Lie group contains all “permutations” of solution curves to an ODE that preserves its solvability (i.e. the reduction of order is possible). In both contexts, the equation is solvable if and only if the associated symmetry group is solvable. Lastly, it should be carefully noted that in both contexts, an equation being “unsolvable” *does not* mean no solutions exist. Just as one can find roots to a quintic with the help of computational softwares, one can locate particular solutions to an ODE of any order in a similar fashion.

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